Distributed Open Science in Practice

A Framework and Proof of Concept from IGDORE

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Introduction

Scientific research is increasingly global, digital, and interdisciplinary, yet its organizational structures remain largely centralized within institutional and geographic boundaries and constraints. This creates barriers to participation, limits inclusiveness, and constrains innovation. This particularly holds for independent, early-career, and geographically dispersed researchers. At the same time, advances in digital collaboration tools, open science practices, and distributed governance models offer new opportunities to reimagine how research can be democratized, organized, and sustained.

In this landscape, the Institute for Globally Distributed Open Research and Education (IGDORE) has emerged as a platform for open and distributed science, providing an institutional framework for independent researchers. However, there is still no widely accepted framework and tested models for how such distributed research networks can be effectively organized, governed, and sustained.

This white paper is the first step in establishing such open, distributed research workflows. The aim is to describe the conceptual and methodological background for distributed open science collaboration. It articulates core principles, such as openness, transparency, inclusiveness, and FAIR data stewardship; proposes governance models, collaboration workflows, and IT tools enabling research without reliance on a centralized institutional core.

Importantly, the document itself is the product of a small-scale proof of concept (PoC), in which a geographically distributed consortium collaboratively wrote, reviewed, and refined the draft. The PoC tests practical workflows, decision-making structures, authorship models, and tool ecosystems, generating insights into both the opportunities and limitations of distributed collaboration.

The result is a set of best practices, governance templates, and implementation guidelines that can inform not only IGDORE researchers but also large-scale international research initiatives, including COST Actions and larger consortia with the potential to attract Horizon Europe consortia. By combining conceptual innovation with practical experimentation, this work contributes to developing a sustainable, inclusive, and trustworthy model for distributed research in the 21st century.

1 The Problem: Structural Limits of Institution-Centered Research Systems

Brief History Research Funding: How Modern Research Funding Took Its Present Form

The organization and funding of research are not merely administrative arrangements; they fundamentally shape which questions are asked, who can ask them, and how knowledge is produced, validated, and disseminated. Historically, scientific inquiry was largely the domain of privileged individuals operating within a patronage system. Early scientists either self-funded their work or relied on wealthy sponsors, including aristocrats, religious institutions, or early industrial actors. While this model enabled foundational discoveries, it also meant that research agendas were often influenced, explicitly or implicitly, by the priorities of patrons (Ziman, 2000).

By the late 19th and early 20th centuries, research began to be institutionalized within universities, industrial laboratories, and scientific societies. However, the most significant transformation occurred during and after World War II. Wartime scientific efforts—such as radar development and the Manhattan Project—demonstrated the strategic importance of large-scale, state-funded research. This led to a reconfiguration of the scientific enterprise into a centralized, government-supported system.

A defining moment in this transition was Vannevar Bush's 1945 report, *Science: The Endless Frontier*, which articulated a vision for sustained federal investment in scientific research in the United States. Bush proposed a model in which public funds would be allocated to universities and research institutions, which would then distribute resources to individual researchers through competitive grants. This framework established the idea of research universities, the foundation for modern research ecosystems, characterized by centralized funding flows, institutional affiliation as a prerequisite for participation, and peer review as a mechanism of quality control (Bush, 1945).

Over time, this model was adopted globally, becoming the dominant paradigm for organizing scientific work. It enabled the expansion of research capacity, the professionalization of science, and the development of robust infrastructure. However, it also entrenched a system in which access to resources, legitimacy, and visibility became closely tied to institutional structures. What emerged was not merely a funding mechanism but a socio-technical system that continues to shape global knowledge production.

Centralized vs. Distributed Research Models

Modern scientific research is largely organized around centralized institutions, including universities, national research institutes, and formal publishing systems. These institution-centered models have historically played a critical role in advancing knowledge by enabling coordination, maintaining standards of rigor, and providing stable infrastructure. Centralization facilitates resource pooling, long-term investment in infrastructure, and regulatory oversight necessary for ethical and safe research practices, particularly in sensitive domains such as clinical trials and biomedical research (Merton, 1973).

Institutional science also provides important protections for researchers. Affiliation offers access to funding, training, mentorship, and legal and ethical frameworks. Institutional review boards (IRBs), data governance systems, and professional norms help ensure accountability, participant protection, and research integrity. In this sense, centralized systems have been essential in building trust in science as a public good.

However, these systems were largely designed in a pre-digital, geographically bounded era. As such, they increasingly struggle to accommodate the scale, speed, and distributed nature of contemporary scientific challenges. Centralization often results in hierarchical control over resources and decision-making, limiting flexibility and responsiveness.

In contrast, distributed research models leverage digital infrastructure, open collaboration, and networked participation to enable more decentralized knowledge production. These models allow researchers to collaborate across institutional, national, and disciplinary boundaries without requiring formal affiliation. Examples include open-source software communities, citizen science initiatives, and decentralized data-sharing platforms.

While distributed models offer increased inclusivity, flexibility, and scalability, they also introduce challenges related to coordination, quality assurance, and sustainability. Without centralized oversight, ensuring standards of rigor, attribution, and ethical compliance can be more complex. Thus, the tension between centralized and distributed models is not binary but reflects a need to balance structure with openness.

A key limitation of centralized systems is the creation of access barriers. Participation is often contingent upon institutional affiliation, access to funding, and integration into established networks. This excludes a growing population of independent researchers, early-career scientists, and those in under-resourced environments. As a result, the benefits of centralized coordination are accompanied by systemic inequities in who can contribute to and benefit from scientific progress.

Trust, Transparency, Integrity, Conduct, and Reproducibility Challenges

Despite their emphasis on rigor and standardization, institution-centered research systems face persistent challenges related to trust, transparency, and reproducibility. Over the past decade, concerns about a “reproducibility crisis” have emerged across multiple disciplines, including psychology, biomedicine, and economics (Baker, 2016). Studies have shown that a substantial proportion of published findings cannot be reliably reproduced, raising questions about the robustness of scientific knowledge.

Several structural factors contribute to these challenges. First, the traditional publishing system often prioritizes novel and statistically significant results over replication studies or negative findings. This creates publication bias and distorts the scientific record. Second, limited access to underlying data, code, and methodologies restricts the ability of other researchers to verify and build upon existing work.

The peer review process, while central to quality assurance, is often opaque and inconsistently applied. Reviewers are typically anonymous, and decision-making processes lack transparency. This can introduce bias, reduce accountability, and limit opportunities for

constructive feedback (Smith, 2006). Furthermore, the concentration of publishing power among a small number of commercial entities exacerbates these issues by reinforcing closed access models and profit-driven incentives (Larivière et al., 2015).

Research integrity is also influenced by incentive structures within academia. The “publish or perish” culture emphasizes quantitative metrics such as publication counts, journal impact factors, and grant income. As highlighted in *The Metric Tide* (Wilsdon et al., 2015), these metrics can distort research behavior, encouraging practices such as selective reporting, p-hacking, and minimal publishable units. Such dynamics undermine the credibility and reliability of scientific outputs.

Transparency initiatives, including open data, open methods, and pre-registration, have gained traction as potential solutions. However, their adoption remains uneven, often limited by institutional policies, resource constraints, and cultural resistance. This highlights the need for systemic change rather than isolated interventions.

The Growing Role of Independent, Early-Career, and Geographically Distributed Researchers

The global research landscape is undergoing a significant demographic and structural shift. Advances in digital technologies, increased access to online learning resources, and the proliferation of open data have enabled a growing number of individuals to engage in scientific research outside traditional institutional settings.

Independent researchers, early-career scientists, and geographically distributed contributors are increasingly participating in knowledge production. This includes researchers in low- and middle-income countries (LMICs), interdisciplinary practitioners, and contributors from non-academic sectors. These actors bring diverse perspectives, local knowledge, and innovative approaches that are often underrepresented in traditional systems.

However, despite their potential, these groups face substantial structural barriers. Institutional affiliation remains a key determinant of access to funding, publication opportunities, and professional recognition. Many grant mechanisms explicitly require institutional backing, effectively excluding independent researchers. Similarly, publication systems often rely on institutional email addresses, affiliations, and funding acknowledgments as markers of legitimacy.

Early-career researchers are particularly affected by these dynamics. They operate within competitive environments where career progression depends on meeting narrow performance metrics. This limits their ability to engage in open, collaborative, or interdisciplinary work, which may not be immediately rewarded.

Geographically distributed researchers, particularly in LMICs, face additional challenges related to infrastructure, funding, and visibility. These constraints contribute to a global imbalance in knowledge production, where research outputs are disproportionately concentrated in high-income countries (UNESCO, 2021).

At the same time, emerging models of collaboration—such as open science communities, distributed research networks, and virtual labs—demonstrate the potential to integrate these

diverse contributors. These models challenge the assumption that high-quality research must be anchored within formal institutions.

Access, Accessibility, and Participation Challenges

Access to scientific knowledge and participation in research systems remains highly unequal. A significant portion of the global scientific literature is locked behind paywalls, limiting access for researchers, practitioners, and policymakers without institutional subscriptions. Article processing charges (APCs) for open access publishing can be prohibitively expensive, particularly for researchers in LMICs.

These financial barriers are compounded by infrastructural constraints. Reliable internet access, computational resources, and laboratory facilities are unevenly distributed globally. In data-intensive fields such as genomics and bioinformatics, the lack of high-performance computing infrastructure can significantly limit research capacity.

Participation in global research networks is also constrained by geographic, linguistic, and socio-economic factors. International conferences, which play a key role in knowledge exchange and collaboration, are often inaccessible due to travel costs and visa restrictions. Language barriers further limit the visibility and dissemination of research conducted in non-English contexts.

These challenges contribute to what has been described as “epistemic inequality,” where certain forms of knowledge and certain groups of researchers are systematically marginalized (Chan et al., 2020). As a result, global research agendas may fail to address locally relevant issues, particularly in areas such as public health, environmental sustainability, and social development.

Efforts to address these challenges have included the promotion of open access publishing, capacity-building initiatives, and international collaborations. Organizations such as UNESCO have advocated for open science as a means of enhancing inclusivity and equity. However, many of these efforts operate within the constraints of existing institutional frameworks, limiting their transformative potential.

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2 Conceptual Framework for Distributed Open Science

Distributed Science and Open Science

Distributed science

Distributed science is not a new idea, even if its current institutional forms are. The *Republic of Letters* — the pan-European network of scholars, natural philosophers, and intellectuals that flourished from the late seventeenth century through the Enlightenment — demonstrated that rigorous intellectual exchange is feasible across national boundaries without centralized institutional control. From Leiden to London to Lisbon, correspondents shared their findings, challenged one another's reasoning, and built a cumulative body of knowledge through letters (then handwritten) and the nascent journals of learned societies, long before the research university assumed its present dominant role. While the infrastructure was minimal, the commitment to shared standards of inquiry was not.

In its contemporary form, distributed science operates on a recognizably similar principle, though with incomparably more powerful tools at its disposal. It is a model of collaborative research in which scientific work is conducted by geographically dispersed individuals and teams, without dependence on a single centralized institutional structure. Coordination is not through physical co-location but through shared governance frameworks, open research practices, and digital infrastructure that makes meaningful collaboration across time zones and national boundaries routine rather than exceptional. In its most developed form, this model supports what might be called borderless affiliation: a researcher's professional standing, institutional identity, and access to the resources necessary for scientific work are decoupled from employment at any particular location. The question the Republic of Letters answered — whether serious scholarship could thrive outside fixed institutional walls — has not changed. The means available to answer it again, affirmatively and at scale, are considerably greater.

Open science

Open science is the movement aiming to make all aspects of scientific research, including publications, data, software, physical samples, and underlying models, transparent and accessible to everyone. It covers a wide variety of practices, such as publishing research under open licenses, making datasets and analysis code publicly available, taking an open-notebook approach to expose the research process rather than just its conclusions, and engaging non-specialist audiences to bridge the gap between scientific knowledge and public life.

The case for openness is based on more than just the principle of accessibility, although this would be sufficient justification for publicly funded research alone. Open practices also address some of the most significant technical limitations of conventional scientific communication. When data and code are shared alongside findings, analyses can be independently verified and errors detected before they propagate through the literature. When methods are documented in sufficient detail for independent replication, the reliability of reported results can be empirically tested rather than assumed. Furthermore, when the complete record of a research project — including null results and unsuccessful approaches — is accessible to the scientific community, the overall view of a field is less distorted by selective reporting. In this sense, openness is not merely an ethical commitment to

transparency, but also a methodological commitment to producing knowledge that is more likely to be true.

Open source software

TODO

On the role of openness

Building meaningful science and "*standing on the shoulders of giants*" depends on a simple but demanding promise, namely that it is possible to find, access, and verify previously reported scientific results. In this context, scientific writings, and even more so publications, connect researchers in a documented chain.

Why are we qualifying science as "*open*"? – Private research has always existed and can be absolutely legitimate. It is a driving force in economic competition, a key strength of the free market. In such a context, research is actually "*closed*", as opposed to "*open*". In the context of research conducted by public institutions, ethical requirements have in recent decades been compressed by economic/political pressures. The "*free market competition*" model has gradually been adopted in the attempt to optimize financial resources, also in the public sphere, at the expense of collaboration. Naturally, these trends hinder accessibility and verifiability of research. So, what we are actually facing in the last few decades is the need to re-open scientific research. Beyond the obvious ethical consideration that public funds are not meant to end up financing private-like 'closed science', there are very relevant practical reasons for putting pressure in this direction: for scientific research, collaboration is crucial.

Technological development provides many enlightening stories of "*open*" overcoming "*closed*". The sharing of software source code had existed before, but the shift toward proprietary licensing by the 1980s prompted the need for more structured approaches, branded "*open source*" with the founding of the Open Source Initiative in 1998. By the late 2010s, the R open source statistical environment surpassed proprietary SAS(R) in scholarly use. Microsoft, at the time, openly described 'open source software' as dangerous and even "un-American", but then pursued a strategic turnaround that culminated in the acquisition of GitHub by 2018! By the early 2020s, QGIS had substituted ArcGIS in universities and for cost-limited practitioners. After a long, slow ascent (the project began in 1986), the migration from Oracle to PostgreSQL reached a tipping point around 2024-2025. Even at the operating system level, open source software holds prominent positions.

Nonetheless, due to the lack of public and academic engagement in sustaining the freedom and effectiveness that "*open source*" entails, private information technology firms are currently ruling the world with an influence that surpasses that of any single state. We can no longer exist without the cloud and, more recently, artificial intelligence, which are mostly proprietary software.

Obviously, proprietary software is valuable and, by no means, should be deprecated. It rather breaks down to a question of proportions: proprietary software in an open environment can provide specific advantages, open software (as well as open science) in a closed environment can end up in a form of self-deception. Some institutions have invested financial and human resources in regaining actual control over their data and in owning the cloud and AI infrastructure, thanks to open source resources (France has recently taken a further step in this direction). In such environments, new and specific tools become available, offering also relevant alternatives to (apparently) freely available proprietary tools.

Obviously, from a practical perspective, the introduction of new tools has an impact on the production chain as an investment cost, but sensitivity and awareness of the underlying implications greatly supersede the value of such costs.

Core principles of openness

TODO

- transparency,
- reproducibility, Diversity, Equity, and Inclusion (DEI), FAIR data, Honest Science, Freedom to science, CARE, TRUST, sharingFrom COS:
    - Transparency: Making research visible
    - Sharing: Making research accessible and usable
    - Inclusivity: Involving and crediting more contributors to research
  - From UNESCO:
    - Transparency, scrutiny, critique, and reproducibility
      - to reinforce the rigor of scientific results, enhance the positive impact of science on society, and increase society's ability to solve complex interconnected problems.
    - Equality of opportunities
      - to ensure that all scientists and those with an interest in science have equal opportunity to access, contribute to, and benefit from science, regardless of origin or circumstance.
    - Responsibility, respect, and accountability
      - to be responsible for and aware of public accountability, potential conflicts of interest, intellectual integrity, and the possible social or ecological consequences of research activities.
    - Collaboration, participation, and inclusion
      - to ensure that scientific collaborations transcend the boundaries of geography, language, and resources, and include knowledge from marginalized communities to solve problems of great social importance.
    - Flexibility
      - to acknowledge that there is no one-size-fits-all way to practice open science and to encourage different pathways to practicing it while upholding the core values.
    - Sustainability
      - to be as efficient and impactful as possible by building on long-term practices, services, infrastructures, and funding models to ensure participation of scientists from less-privileged countries or institutions.

Honest Science: A Commitment to Meaningful Inquiry

Honest Science is a principle centered on the integrity of the researcher's intent.

We strive, to the best of our ability, to be honest with ourselves and with others, specifically in two aspects:

(1) choosing research problems that we genuinely believe to be important; and
(2) reporting our results in a way that we genuinely believe to be accurate.

1) Honesty in problem choice
requires us to examine the true motivations behind our work. Are we pursuing questions because they are fashionable, safe, or convenient, or because we have a deep-seated belief that they are important? This principle moves beyond the avoidance of data falsification to address a more subtle form of dishonesty—working on problems we do not fundamentally care about.

- *Hamming, R. W. (1986). "You and Your Research." Bell Communications Research Colloquium Seminar.*
- *Liska, P. (2004). "Viewpoint: The morality of problem selection in proteomics." Proteomics, 4(7), 1957-1960*
- *Baehr, J. (2011). The Inquiring Mind: On Intellectual Virtues and Virtue Epistemology. Oxford University Press. (Specifically Chapter 9 on Intellectual Courage).*

2) Honesty in reporting
demands we resist the pressure to overstate or understate what we genuinely believe about our findings, or present results with a certainty we do not feel. We commit to communication that reflects our true beliefs about the work's genuine importance, achievement, and limitations.

- *Feynman, R. P. (1974). "Cargo Cult Science." Engineering and Science, 37(7), 10-13*
- *Dormandy, K. (2020). "Intellectual Humility and Epistemic Trust." In The Routledge Handbook of Philosophy of Humility. Routledge.*

Honest Science, therefore, is a commitment to "epistemic debugging," where we actively examine how external pressures and unexamined assumptions have shaped our research choices. It holds that the most robust and valuable science emerges when we are willing to update not just our hypotheses, but our research priorities, in response to what we genuinely find meaningful and true. It is a dedication to ensuring our professional output aligns with our intellectual and ethical convictions.

TODO

3 Recommended Practices for Distributed Open Science Collaboration

This set of recommended practices for distributed (e.g., remote) collaboration represents the experiences and opinions of the document's authors. Where indicated, these practices include templates and assessment frameworks that will be periodically reviewed for refinement into evidence-based best practices. Teams and institutions that adopt these practices are encouraged to share their assessment data in order to improve collaborative efforts throughout the globe and across disciplines.

Collaboration workflow template

Open science has long been a grassroots movement, led by motivated individuals who contribute their considerable expertise and passion to collaborative projects. This is our heritage as OS practitioners, and as our movement has matured and merged with mainstream sponsored research, this is an opportune time to establish processes and standards that honor our tradition of volunteerism while improving efficiency and outputs, easing collaboration and interdisciplinarity, and increasing competitiveness for awards.

All collaborative efforts require leadership, roles, tasks, deadlines, milestones, and oversight--particularly complex research projects with several stages that will take place over an extended timeline. The bystander effect notwithstanding, it is the responsibility of the project's leadership to select a team, assign roles and tasks, actively manage the workflow, and oversee the production of data and knowledge outputs. The contributions of OS talents are not to be exploited or taken for granted–it is a precious resource that deserves the proper support to see that expertise and effort through to the delivery of works.

In practice, this means that OS initiatives should plan for processes, accountability mechanisms, and administrative interventions that roughly parallel a university-hosted project. For example, projects should have a lead investigator and a project manager, which may be one and the same, but which obligate the person(s) to complete the OS checklist (see above) by taking ownership of the activity. Although OS projects accommodate flexible and inclusive practices, research team members should be vetted, assigned a specific role with a thorough and thoughtful "job" description, and given tasks that are situated within the overall workflow. In other words, a "do what you can, as you can" work ethic should not be tolerated on research projects, as it undermines the urgent effort that others contribute and demoralizes the team over time. Instead, a proactive plan that ensures the completion of each task should be designed at the outset and implemented throughout the project lifecycle.

Finally, while OS work encourages interdisciplinarity and active input from the larger community, the priority should always be on producing and delivering new tools, processes, and knowledge. This means that activity leaders should acknowledge and accommodate the separate processes of (1) culture building, which explores community standards and values in an indefinite continuous state; and (2) knowledge production, which is the result of structured collaboration defined by output commitments and deadlines.

Workflow Roadmap

Ideation: this step involves sharing a new idea. Using the community's established communications platform, On Science and Academia, a scholar should announce their

interest in exploring a new idea, including references to published research, funding opportunities, and/or potential partners. This step allows the OS community to begin self-selecting for a potential role as an investigator, advisor, reviewer, administrator, or contributor to the effort. The original poster should clarify their preferred role should the idea advance to the proposal stage.

Information gathering: this step sparks the first scientific investigation into an idea that the community has decided to move forward on. This preliminary activity should identify scholarship, existing groups working on similar activities (Birds of a Feather), conferences and events that IGDORE might support travel for, potential partners and investigators, and relevant literature.

Project design moves the initiative forward by developing a preliminary research plan that can be easily translated into grant proposal elements. This step involves team building, literature review, methodology, hypothesis, identification of available & needed facilities and equipment, and a justification for the effort. At this stage, project leaders should prepare for a long-term engagement with a significant time commitment–particularly for unfunded work, which may require more intensive oversight and micromanagement of certain tasks throughout the project lifecycle.

Proposal development further advances the initiative by assigning a project manager (PM) or research administrator (RA) to work closely with the principal investigator (PI) to formalize the proposal for grant writing or grassroots organization purposes. This step adds elements to the project design and refines operational details, including affiliation, budget development & justification, data collection and management plan, expected output, and additional documentation.

Project management planning: this step should take place at the onset of the activity, either as a sponsored project with external funding or as a volunteer effort reliant on unpaid talent and facilities.

Assessment Framework (Closeout Report)

This simplified metrical framework measures the effectiveness of recommended practices by capturing the adoption of discrete recommendations and assigning a binary value to each, which is then calculated as a confidence interval. Even small sample sizes will produce meaningful results, which will directly impact the delivery of evidence-based best practices over time. It should be noted that these practices apply to IGDORE activities, and may not translate well outside of OS and/or distributed research contexts.

Did you apply the following recommended practice to your IGDORE activity?	**NO**	**YES, and I recommend this practice**	**YES, and I do not recommend this practice**
Host a Birds of a Feather event to introduce and explore a potential WG, IG, or COP activity	0	+1	-1
Assemble interested Scholars to co-create objectives, timeline, roles, and tasks	0	+1	-1
Register the activity with IGDORE	0	+1	-1

Create a Gantt chart, project matrix, or other project tracking tool	0	+1	-1
Commit to and execute assigned tasks	0	+1	-1
Notify IGDORE when the activity has ended	0	+1	-1
FAIRify Data collection and management plan	0	+1	-1
Ensure that data about people meets CARE standards	0	+1	-1
Register the data management plan on OA platform	0	+1	-1
Preregister project design, hypotheses, and details on OA platform	0	+1	-1
Publish project findings on an OA platform	0	+1	-1

Governance and contribution template

Open science collaboration is deeply grounded in ethical idealism–cooperation, fairness, equality, hope, the right to pursue knowledge, the freedom to pursue diverse and divergent ideas, and the core belief that scientific advancement should be based on merit instead of political or economic considerations. The OS community of practice represents the purest values and boasts the most passionate researchers, who frequently work on their own time and at their own expense to advance scientific knowledge for the benefit of the entire world. In this context, distributed workflows for OS collaboration should be considered.

It is a privilege to work with the best and brightest, and they deserve all the professional support and tools that IGDORE can offer, without imposing an undue burden on volunteer talent. In a university, government, or industry lab, research is conducted with administrative support to coordinate meetings and events, handle payments and costs, set deadlines and track milestones, and ensure that the entire process runs smoothly. In contrast, volunteer efforts tend to be hands-off affairs that fall on a handful of people (and frequently a single individual) to follow through. Both approaches have their costs and benefits. Professionalized projects can become an expensive, bloated affair with micromanagers that demoralize the research staff. Meanwhile, volunteer projects that lack firm leadership can lose momentum and focus, ultimately creating a lot of busywork with no meaningful output. This section outlines an approach that strikes a balance between managed projects and ad hoc efforts through structure (Birds of a Feather, Working Group, Interest Group, and Community of Practice), roles (project manager or owner, researcher or contributor, and advisor or reviewer), and oversight mechanisms (assigned tasks, deadlines, milestones, and shadowing). A deeper understanding of these dynamics will help IGDORE Scholars better align their expectations for collaborative experiences. This will empower them to contribute to a strong, uniquely IGDORE approach to open science that leads the global community.

Governance Structures

The first task in designing a workflow is to identify the objective of the activity. With few exceptions, most decentralized collaborative efforts will fall into one of three categories:

1. **Birds of a Feather** (BOF), which introduces a potential activity by announcing it to the general membership to determine interest and recruit potential members;
2. **Working Group** (WG), which aims to produce a pre-defined output, such as knowledge, literature, curriculum, code, technology, or an event (e.g., conference, meeting, etc.);
3. **Interest Group** (IG), which aims to investigate and discuss topics of concern, urgency, or general interest that may result in a formal assessment, a set of recommendations, or lead to the creation of a Working Group that undertakes a new project with deliverables; and
4. **Community of Practice** (COP), which aims to foster relationships between members and share knowledge by way of blogging, chatting, virtual brown bags, and cocktail hours, and more formal symposia, like lightning talks and sharing papers.

It is further helpful to assess the capacity of the core group participating in certain activities. For example, a few motivated Scholars may have the skills and wherewithal to move forward on a project, but are unable to confirm a regular commitment to participate and meet deadlines. In that case, creating an Interest Group might be a more realistic goal than a Working Group, which might ultimately frustrate members who contribute a significant effort that fails to culminate in a deliverable. On the other hand, a long-standing Interest Group can mobilize into an active Working Group with the introduction of new talent, funding, or other opportunities to move forward.

Roles

It is the current recommended practice that each type of workflow must be registered with appointments and assigned tasks and responsibilities in each of the three roles:

1. **Lead**: This person(s) assume ownership of the activity's objective. This may involve hosting a virtual meeting on occasion, as with a COP liaison; it might require somewhat more effort, such as an IG coordinator leading an institutional assessment; or it could be quite an intensive experience, as would be expected of a WG chair managing a project whose objective is to reproduce findings and publish results.
2. **Investigator**: These Scholars conduct lab, field, and/or desktop research and/or directly contribute to research objectives. This role is primarily involved in gathering and reviewing information (e.g., literature review and commentary); formulating hypotheses; designing and executing methodologies; collecting, managing, and assessing data; and contextualizing findings and knowledge. Working Group investigators are generally research scientists and tech developers. Interest Group and Community of Practice investigators contribute expertise to activities.
3. **Peer**: These IGDORE Scholars give expert support to activities as advisors and reviewers. This role may range from passive, such as mediation in case of conflict, to very active, such as validating citations and data in preparation for publication. Peers in a Working Group might advise on project design, recommend team members, and/or provide insight at various stages of the project. In an IG or COP setting, they may be asked to make a special contribution based on their experience and expertise.

All of these basic roles ensure that our Scholars' voluntary efforts are treated with the respect and seriousness that is expected of a serious society of scientists and academics. Each type of role should contribute its own kind of support to the individual activity, while ensuring that IGDORE's overall effort is making an important contribution to the advancement of open science as a practice.

The most important operational role in any collaborative effort is the lead, who sets the direction and tone of the work. This might be the originator of the idea, e.g., original poster (OP) of an On Science and Academia blog post, a Community of Practice **liaison**, an Interest Group **coordinator**, or a Working Group **chair**. For complex and long-term projects, it may make sense to distribute the workload across multiple vice-chairs, although it is the current recommended practice to vest a single lead with responsibility for overseeing the project as a whole. This approach simplifies the management of practical tasks, such as a point of contact, scheduling meetings, managing permissions, notifying the team of important updates, etc.

The most important production roles are led by investigators and contributors, who conduct research, experiments, data collection, and data analysis. They are also responsible for producing publishable works, including papers, code, and multimedia products.

Peers provide oversight, guidance throughout the project lifecycle, advice, and feedback on scientific questions and non-scientific issues, such as interpersonal conflicts or management needs. Reviewers may assist with preparing and editing the work.

Oversight Mechanisms

IGDORE aspires to be the global leader in volunteer-led OS research. But all scientific research is a highly complex effort that greatly benefits from some basic administrative support and management approaches. The following oversight mechanisms should be executed by Chairs, Coordinators, and/or Liaisons in a fully transparent manner that allows all members to quickly and easily orient themselves in the project.

- **Assigned tasks**: The work of the project should be delegated to the team with clearly defined tasks. For example, a clearly defined task is, "*Gustav, please write 2 pages on the history of AI development at Google between 2020 and 2025*," while a poorly defined task is, "*Somebody, please write something up about AI and Google*".
- **Deadlines**: A deadline should be set for each task, and it is very helpful to set up a matrix of tasks, deadlines, and the responsible person. A Gantt chart works this way, but a simple table serves the same purpose for many activities. The Lead should provide a clear process for managing deadlines. For example, a good process is to (1) poll the group for a sense of availability and realistic deadline schedule; (2) using transparent poll results, set a sensible timeline of tasks with deadlines and welcome offline comments or concerns to accommodate folks with a conflict; (3) individually check in with each contributor or vice-chair about halfway to the deadline to ask about their progress and ask if they need support to meet their commitment; (4) devise an intervention or support mechanism for tasks that are hopelessly behind schedule, such as re-assignment or personal assistance; and (5) track progress with a visual tool, such as a Gantt chart or table, that everybody can see. This lets members know when to pitch in when an overdue task needs to be completed, and might motivate others to stay on top of deadlines. Whatever process is employed, it should never be used as a negative incentive; all free effort is invaluable, and many

give their time despite having myriad other demands on their time. A good Lead is helpful and supportive and able to keep multiple tasks on target and on time.

- **Milestones**: Milestones are very important in letting the team assess their progress and adjust accordingly. A milestone might be marked by the end of data collection, registration on a pre-print platform, securing research funds, completing a draft version of a paper, or an invitation to partner with another team or institution.
- **Shadowing**: IGDORE can model an innovative approach to oversight and management through shadowing. In this process, each role is assigned to two IGDORE Scholars, with one assuming the main role and the second willing to step in and support as needed. This novel tactic may increase camaraderie between Scholars and improve a sense of security and stability within the project team. Leads may opt to include this in Step 4 of the process recommended above.

Fostering Serendipity

Serendipity in the scientific context is the idealized outcome of knowledge transmission. Simply put, serendipity is the act of stumbling upon an answer to a question that you didn't even know to ask. The ideal case scenario results from researchers swapping information and ideas in a way that sparks innovation–the realization that a process or tool in one discipline would solve a persistent problem in another discipline, the co-production of a genuinely new idea, the creation of a new partnership that leads to scientific discovery, and other activities that deeply motivate new lines of inquiry.

Unfortunately, the highly structured format of virtual meetings generally prevents attendees from interacting in real time, while etiquette discourages meaningful discussion in the public chatroom, relegating potentially valuable leads to private and offline conversations. The convenience of meeting online undermines the productive potential created by interpersonal interactions. However, an intentional meeting management style can offset this unintentional harm.

In practice, serendipity can be fostered by allowing work meetings to flow in an unstructured manner, even at the expense of sacrificing agenda items. For example, when a flurry of chat erupts in the sidebar, it might be a good opportunity to give space to that conversation and welcome the discussion to continue on the monitor–and perhaps to continue beyond the scheduled end time of the event for anybody who wishes to stay online. This method might seem a bit jarring to those who value structure and efficiency, but announcing this policy in the meeting invite will prepare attendees for the possibility.

Governance Roadmap

Use this roadmap to plan an IGDORE activity. Leads should communicate this roadmap to their teams as IGDORE's standardized workflow for distributed open science.

☒Host a Birds of a Feather event to introduce and explore a potential WG, IG, or COP activity
☒Assemble interested Scholars to co-create objectives, timeline, roles, and tasks
☒Register the IGDORE activity
☒Create a Gantt chart, project matrix, or other project tracking tool
☒Commit to and execute assigned tasks
☒Notify IGDORE when the activity has ended
☒Complete closeout report

Project Matrix

A table like the one below can help track project tasks, timelines, and progress. An up-to-date open access tool should be available to the project team throughout the project lifecycle.

Deadline	Task	Assigned To	Stage
Jun 1, 2026	Proposal development	Amber	Complete
Oct 2, 2026	Data analysis	Ruby	In progress
Nov 4, 2026	Literature review	Crystal	Not started

Registration Template

Use the form template below to register your IGDORE activity. After review, it will be posted to the IGDORE website. Members and materials will be published for open access to the public.

Activity Name

Activity Type (choose one)
☒Community of Practice ☒Interest Group ☒Working Group

Summary

- *Objective*
- *Anticipated timeline (WGs & IGs only)*
- *Summary of activities (750 words or fewer)*

Roles

Lead(s)

- *List all activity leads; please edit if the lead changes*

Investigator(s)

- *List all investigators; please add new members as they join the group*

Peer(s)

- *List all peers; please add new members as they join the group*

Tasks & Assignments

Lead(s) Taskset

- *Which tasks will the Chair, Coordinator, or Liaison perform to meet activity objective(s)?*

Investigator Tasks & Assignments

- *Summarize tasks and members assigned to them, including shadow when applicable*

Peer Tasks & Assignments

- *Summarize assigned peer tasks and/or shadows*

Open Science Tools & Practices

- *List the open science tools, methods, and platforms that you will use to host this activity; please include DOIs and update as necessary*

Data, transparency, and documentation checklist

Data Stewardship

The most valuable and enduring research asset is data. When properly curated, data can be reused. Naturally, each time it is reused, it gains value. Recognized good practices in open science data stewardship begin with the FAIR[1] principles:

- **F**indable: state-of-the-art tools and techniques should be applied to the collection, management, and storage of research data so that current and future investigators can easily locate (find) and access (use) that data for their own research. This practice supports the OS principles of reproducibility, transparency, and sharing.
- **A**ccessible: data should be stored on platforms that prioritize data sharing. This might include paid access, subscription services, or free options. The goal is to ensure that data is released from private servers, stored hard drives, and other media that cannot reasonably be attained by an outside investigator. This practice supports the OS principles of sharing, equality, participation, and inclusion.
- **I**nteroperable: data and metadata should be managed for use across different languages, operating systems, data management platforms, and formats (e.g., visual, audio, non-tabular, textual, etc.). This practice supports the OS principles of flexibility, inclusion, equality, and reproducibility.
- **R**eusable: rich metadata should be assigned to every dataset to ensure that restrictions, licenses, and provisions are clearly defined, provenance and access are properly managed, and long-term storage is ensured. This practice supports the OS principles of sustainability, inclusion, sharing, reproducibility, responsibility, and participation.

When considering how to develop a project's data stewardship design and Data Management Plan commitments, the Beijing Declaration on Research Data[2] offers clear guidance on why, when, and how to share research data in a global context, including the treatment of restricted, classified, and sensitive data.

Some OS collaborations might require a more specialized set of good practices to ensure that the human rights, privacy, and heritage of individuals and communities are protected, even while making research data as open as possible and methodology as transparent as possible. The recommended starting point is the Indigenous Data Sovereignty and Governance (IDS) framework[3], which may be broadly applied to any social research design, as well as other disciplinary projects that may capture environmental data about people, such as traffic studies that captures personally identifiable information (PII) through license plates and traffic cams, audio and video files collected from wildlife studies that capture

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[1] Wilkinson, M., Dumontier, M., Aalbersberg, I. et al. The FAIR Guiding Principles for scientific data management and stewardship. Sci Data 3, 160018 (2016). https://doi.org/10.1038/sdata.2016.18.

[2] CODATA, Committee on Data of the International Science Council, CODATA International Data Policy Committee, CODATA and CODATA China High-level International Meeting on Open Research Data Policy and Practice, Simon Hodson, Barend Mons, Paul Uhlir, and Lili Zhang. "The Beijing Declaration on Research Data". Zenodo, November 25, 2019. https://doi.org/10.5281/zenodo.3552330.

[3] Kukutai, T. and Taylor, J. Indigenous data sovereignty: Toward an agenda. 2016. Canberra, Australia: Australian National University Press. DOI: https://fnigc.ca/wp-content/uploads/2020/09/bbe195ddc231e3b1222d71ca4c09ae62_indigenous_data_sovereignty_toward_an_agenda_11_2016.pdf.

voices and images of people in the vicinity, etc. IDS represents a call to thoughtful caution about norms and sensitivities when collecting and curating data about people and cultures, and is expressed through the CARE Principles for Indigenous Data Governance[4]:

- **C**ollective benefit: data stewardship and project design should incorporate inclusive development and innovation, lead to improved governance and community engagement, and produce equitable outcomes. This practice supports the OS principles of inclusion, sharing, responsibility, equality, and participation.
- **A**uthority to control: data stewardship and project design considerations should recognize the rights and interests of the study's subjects and partners, follow the community's data governance protocols, and ensure the involvement of the community in the long-term stewardship of collected data. This practice supports the OS principles of responsibility, participation, flexibility, collaboration, inclusion, respect, and accountability.
- **R**esponsibility: research teams must assume responsibility and establish active oversight mechanisms to ensure a respectful relationship with the community that provides the project's research data, which may extend to technical assistance agreements, training, and/or access to data tools and assets. This practice supports the OS principles of responsibility, collaboration, inclusion, respect, and accountability.
- **E**thics: under the guidance of local stakeholders, the rights and well-being of individuals and communities should be a priority consideration in the design of methods and data stewardship, with particular emphasis on minimizing harm, maximizing benefit, promoting justice, and planning for the future use of data. This practice supports the OS principles of responsibility, scrutiny, participation, flexibility, collaboration, inclusion, respect, and accountability.

Transparency Tools & Standards

Finally, OS projects must deliver transparency, which is ensured through process and oversight in a human-in-the-loop research system. The data science community is a rich resource for good practices in distributed collaboration, because they have a long history of working asynchronously outside the core research team. Many of the recommended practices for data stewardship parlay into good practices for open science collaboration activities writ large. The primary OS output is the publication of results and data, and that process further advances transparency and documentation practices.

Open science outputs are mainly delivered in three ways: (1) publishing findings and results in an open access (OA) medium; (2) reporting project design and hypotheses on a preregistration platform prior to data collection; and (3) submitting the data management plan (DMP) on an OA platform prior to project commencement. Publishing opportunities can be found in a robust landscape of OA platforms, including curated project websites, journals (e.g, Nature Communications, etc.), distributors (e.g., arXiv.org, etc.), collaboration platforms (e.g., github.com, etc.), publishers (e.g., PLOS, etc.), and blogs (e.g., medium.com, etc.). Preregistration is a critical step in conducting an OS activity because it prevents both intentional and inadvertent ethical violations related to data manipulation (i.e., cherry picking, obfuscating negative findings, failing to report data, etc.), while advancing transparency by

[4] Carroll, S, et al. 2020. The CARE Principles for IndigenousData Governance.Data Science Journal, 19: XX, pp. 1–12. DOI: https://doi.org/10.5334/dsj-2020-042.

reporting the methods, tools, team members, and sponsor details at the outset of the project. The Center for Open Science offers training, information, storage, and templates for preregistration. The DMP delivers the most valuable OS assets–research data–by publishing the stewardship plan to collect, curate, and share that data over the long term, including post-closeout management and deep storage.

The Data Management Planning Tool (https://dmptool.org/) is the standard platform for developing and posting OS DMPs, while the TRUST principles[5] offer meaningful guidance in the decision-making process, which is a particularly valuable framework for interdisciplinary teams that come to the project with different standards and expectations. The TRUST framework helps investigators choose the right repository for their data throughout and beyond the project lifecycle by describing the relevant criteria for responsible, trustworthy data stewardship:

- **T**ransparency addresses policies, practices, terms of use, digital preservation timeframe, and stewardship details related to capacity, sensitive data, and deep storage.
- **R**esponsibility addresses metadata and curation standards, machine interfaces, access protocols, authenticity, security, IP and licensing, and accountability.
- **U**ser focus promotes the global research data commons by implementing data metrics for findability and interoperability, and tracks evolving community standards by regularly updating tools, policies, and services.
- **S**ustainability is demonstrated through post-lifecycle planning (i.e., risk mitigation, business continuity, disaster recovery, and succession), stable funding and institutional strategic planning, and governance for the long-term preservation of data so that data resources remain discoverable, accessible, and usable in the future.
- **T**echnology must remain up to date on appropriate standards, tools, and technologies for data management and curation, and publicize its threat mitigation plans and mechanisms.

OS Checklist

- ☐ The data collection and management plan is FAIR
- ☐ Data about people meets CARE standards
- ☐ The data management plan has been registered
- ☐ Preregistration of project design, hypotheses, and details
- ☐ Project findings published on an open access medium

- Pre-project authorship agreement - see CARE framework below https://www.care-council.org/
- Preregistration
    - www.aspredicted.org
    - https://osf.io/
    - https://zenodo.org/
    - https://www.socialscienceregistry.org
    - https://www.3ieimpact.org

[5] Lin, D., Crabtree, J., Dillo, I. et al. The TRUST Principles for digital repositories. Sci Data 7, 144 (2020). https://doi.org/10.1038/s41597-020-0486-7.

- Inclusiveness and gender guidelines

4 Support Structures and Resources

A globally distributed ecosystem of independent scholars requires accessible support structures that both enable collaboration and mitigate the risks commonly associated with academic work, such as authorship disputes, power abuse, harassment, and restrictions on academic freedom. The following resources and frameworks illustrate key pillars of such an infrastructure.

- Authorship agreements and their revisions, e.g., CARE framework: https://www.care-council.org/

  One of the foundational elements of such an ecosystem is the use of clear and adaptable authorship agreements. CARE (Council for Authorship Rights and Ethics) provides educational materials, templates of authorship agreements and their revisions, and guidance on defining authorship criteria, contribution thresholds, assigning roles, and ensuring transparency in how credit is allocated. For distributed teams, these agreements are especially important because collaborators often work across different institutional, cultural, and disciplinary contexts. Rather than being treated as static documents, authorship agreements should be understood as evolving instruments that are revisited throughout the lifecycle of a project. This allows teams to account for changing contributions, incorporate new collaborators, and address potential tensions early, reducing the likelihood of authorship problems.

- Resolving authorship disputes pre- and post-publication, e.g., CARE: https://www.care-council.org/

  Even with agreements, authorship problems may still arise both before and after publication. In such cases, access to independent advisory bodies becomes essential. Initiatives such as CARE can help collaborators navigate disagreements in a structured and neutral manner. Pre-publication mediation is particularly valuable, as it helps prevent reputational harm, delays in dissemination, and the escalation of conflicts. Post-publication disputes require additional mechanisms for transparency and correction, including the possibility of amendments to authorship records and formal documentation of contributions. For independent scholars, these external mechanisms effectively replace institutional channels and provide a necessary layer of oversight. They also serve an important protective function by mitigating potential power imbalances, particularly in collaborations involving partners with greater institutional resources.

- Resources for dealing with academic bullying, harassment, and power abuse, e.g., MaWi, APM: https://www.netzwerk-mawi.de/
  https://paritymovement.org/

  Another critical area of support concerns academic bullying, harassment, and abuses of power. Independent and early-career researchers are often especially vulnerable in collaborative settings, particularly when working with more established or institutionally affiliated scholars. Organizations such as Netzwerk Mutige Wissenschaft (MaWi) and the Academic Parity Movement offer important resources, including confidential advice, peer support, guidance on navigating reporting processes, and advocacy for the victims. They also play a broader role in documenting systemic issues and raising awareness about harmful practices within

academia. Access to such networks helps reduce isolation, empowers individuals to respond to misconduct, and contributes to fostering a research culture grounded in respect and equity.

- Resources for dealing with violations of freedom to science (institutional censorship):
  https://academicwhistleblower.eu/
  https://academicrightswatch.se/
  https://council.science/freedom-and-responsibility-in-science/

  In addition to interpersonal risks, independent scholars may face structural challenges related to the restriction of academic freedom, including censorship, suppression of findings, or retaliation for work on contentious topics. Resources such as the Academic Whistleblower Network, Academic Rights Watch, and the International Science Council provide guidance, advocacy, and monitoring of violations. They offer channels through which scholars can raise concerns, seek support, and contribute to broader efforts to uphold international norms around academic freedom. For researchers operating outside institutional frameworks, these organizations are particularly important in ensuring that intellectual independence is not compromised.

Together, these support structures point toward the need for an integrated and globally accessible ecosystem that combines ethical guidance, conflict resolution, personal protection, and advocacy for academic freedom. Rather than replicating traditional institutional models, such an ecosystem can be modular and decentralized, reflecting the realities of distributed scholarship while maintaining high standards of integrity and accountability.

5 IT Tools Supporting Concepts and Recommended Practices

Introduction

Collaboration has always been vital in scientific research. Recent advances in Information Technology (IT) have exponentially enhanced this framework, significantly increasing the number and types of connections and nearly eliminating time constraints in communication. Online presence enables fast turnarounds, even real-time collaboration, regardless of the number or geography of collaborators. IT has made Distributed Open Science a reality.

Technological progress has shown that open source solutions can surpass proprietary ones, despite the dominance of private IT firms today. The choice of IT tools that support distributed open science in practice is not neutral due to ethical and political implications. Balancing proprietary and open software is crucial, as investing in open source software can help institutions regain control of their resources and address sustainability concerns.

The section is organized by theme loosely following a logical progression. Repositories are basic nodes in the distribution web. AI is here; the issue is why and how to use it. In the context of ‘distributed open science’, more than in others, understanding each other well is vital; metadata and ontologies serve this purpose. Conceptual tools like the ‘data management plan’ and technical offerings like the ‘open science framework’ or preprint and open peer review solutions should be part of recommended practices in research implementation, beyond the specific technical choice of the tool. In conclusion, some consideration should be given to the tools on which we might spend most of our time as development and collaboration environments.

Repositories

Code sharing

Concerning code (and texts or limited dataset) the world reference for open source collaboration (sharing and version control management) is Github, as mentioned, a subsidiary of Microsoft since 2018. As it occurred, the acquisition of GitHub by Microsoft generated a remarkable alarm with a great increment in GitLab accounts but the move was very farsightedly managed: no apparent change could be perceived by the users of Github and its leading position in the open source world community is still uncontrasted. Gitlab, the second largest concentration of open source projects, can be regarded as a ‘recommended practice’ since it is an independent enterprise, it offers the open source ‘Community Edition' and is explicitly designed for self-hosting. Institutions and entities that can afford it and deploy self-hosted GitLab installations.

<<in case the ‘git’ topic is not deepened in other sections, it can be expanded here>>

Data sharing

Concerning the sharing of data and publications many alternatives exist. The main repositories (Table 1) are global cloud resources that mint a DOI for every published item and manage version updating. Zenodo is the reference as public service and Figshare is largely employed, but the many characteristics that have to be considered build up a greatly diversified range. Commercial interest in offering the service is so high that free options are customary but, what really makes the difference between ‘just being accessible’ and ‘being truly findable’ (by an agent that has no info on its existence), is ‘data curation’ and this can

not come without financial and time costs (https://makedatacount.org/read-our-blog/community-perspectives-open-infrastructure/ DOI: 10.60804/MX4V-7K47)

Table 1 - Relevant global data repositories

Platform	Ownership / governance	Publishing Cost	Scope / strength
Zenodo	Public (intergovernmental)	Free	Cross-disciplinary, GitHub-friendly
Figshare	Commercial	Free (1 GB) / Paid plans available	All research outputs, broad disciplines
Harvard Dataverse	Public / Academic	Free	Social/quantitative data is strong curation / optional-paid
Dryad	Non-profit	Paid (~$120 per dataset)	Publication-linked datasets, robust and routine curation
Mendeley Data	Commercial	Free	Research-data-first, cloud-based
OSF	Non-profit	Free	Project-centric, reproducible workflows

For institutions and entities that can afford it, a relevant alternative is provided by the federated repositories model (based on open source software like Dataverse or CKAN -or based on proprietary software from vendors like Palantir) but, as said, the move requires also supporting the financial and human resources costs of acquisition and maintenance of the hardware involved.

Sharing references

The use of 'Reference Management Software' should be included as a recommended practice in research development and collaboration. As stated, scientists build on others' productions, efficiently and effectively managing and sharing this basic information is obviously important. Mendeley (Elsevier) and EndNote (Clarivate) are among the most well known proprietary solutions. Zotero (Corporation for Digital Scholarship) is the open source solution that has gained high popularity and can be included as a recommended practice.

The Collaborative Open Data Exchange (CODEX) protocol is an open, decentralized infrastructure for managing and exchanging scientific knowledge as versioned, persistent, and interoperable research objects. Rather than treating publications as static outputs, CODEX enables the representation of all research components—manuscripts, datasets, code, peer reviews, and metadata—as modular and continuously evolving entities. Each object is assigned a persistent identifier and can be independently updated, linked, and verified, supporting transparent provenance tracking and reproducibility. The protocol is designed to operate across distributed storage and indexing systems, allowing multiple actors to contribute to, validate, and reuse scientific outputs without reliance on a central

authority. This architecture aligns closely with FAIR and open science principles by making research outputs more findable, accessible, interoperable, and reusable, while also enabling new forms of attribution, validation, and collaborative knowledge building.

Built on top of this infrastructure, DeSci Publish provides a user-facing platform that operationalizes CODEX for researchers, editors, and institutions. It offers an intuitive interface for submitting, versioning, reviewing, and publishing research in an open and transparent manner. Unlike traditional publishing systems, DeSci Publish supports continuous peer review, granular attribution of contributions, and the integration of diverse research outputs beyond static PDFs. It also enables automated metadata enrichment, persistent linking of research components, and interoperability with external systems such as DOI registries (CrossRef) and researcher identifiers (ORCID). By combining decentralized infrastructure with accessible tooling, DeSci Publish lowers barriers to participation in global research collaboration and enables distributed teams to coordinate, validate, and disseminate knowledge more efficiently and transparently.

Collaborative Writing Tools

Effective collaboration across distributed teams requires writing environments that ensure authorship transparency, support asynchronous contribution, and produce outputs in formats compatible with open dissemination. Platforms like Google Docs have gained popularity because they are easily accessible and offer a threaded commentary system (Stokel-Walker, 2022). However, privacy-respecting open-source alternatives such as Framapad and Etherpad offer institutional self-hosting, thereby satisfying data governance requirements without sacrificing real-time co-authoring functionality (Framasoft, 2023). For manuscript preparation in technical disciplines, Overleaf remains the de facto LaTeX-based collaborative editor, offering direct submission pipelines to major preprint servers and journals (Overleaf, 2024). Table 1 summarises the major collaborative writing platforms relevant to distributed open science teams.

Table 1. Collaborative writing tools: features and open science relevance.

Tool	Type / Model	Key Features	Open Science Relevance
Google Docs	Proprietary / Cloud	Real-time co-editing, comment threads, version history	Widely accessible; supports asynchronous cross-institutional collaboration
Framapad	Open-source / Self-hostable	Browser-based pad, colour-coded authors, multi-format export	Privacy-respecting; no account required; aligns with open infrastructure principles
Etherpad	Open-source / Self-hostable	Lightweight, plugin-extensible, API support	Full data sovereignty for institutions; integrates with custom workflows

HackMD / CodiMD	Open-source / Cloud or Self-hosted	Markdown-native, slide mode, Git sync	Ideal for technical documentation and reproducible research notes
Overleaf	Freemium / Cloud	LaTeX collaboration, journal submission integration, track-changes	Standard for scientific manuscript preparation; integrates with preprint servers

Data Sharing Tools and Repositories

Open science mandates that research data be Findable, Accessible, Interoperable, and Reusable (FAIR; Wilkinson et al., 2016). Disciplinary and general-purpose repositories differ in the metadata schemas, embargo policies, and curation levels they offer. The Open Science Framework (OSF), maintained by the Center for Open Science, provides an end-to-end project management environment linking preregistrations, data, materials, and manuscripts under a single persistent identifier (Center for Open Science, 2024). European infrastructure such as Zenodo, hosted by CERN, offers unrestricted long-term archiving with automatic DOI minting and native integration with GitHub release workflows — a critical feature for code-centric projects (Zenodo, 2024).

Discipline-specific platforms such as PANGAEA for earth sciences (PANGAEA, 2024) and the Harvard Dataverse network for social sciences (King, 2007) provide richer domain metadata, improving discoverability within research communities. Table 2 presents a comparative overview of key data-sharing repositories.

Table 2. Data sharing repositories: FAIR compliance features and domain focus.

Platform	Domain Focus	FAIR Compliance Features	Notes
OSF (Open Science Framework)	General research	Persistent DOIs, version control, project registry	Flagship platform of the Center for Open Science; integrates with GitHub and Zotero
Zenodo	General / CERN	DOI minting, long-term archiving, GitHub release sync	Free; trusted European infrastructure; widely used for supplementary data and code
Figshare	Multidisciplinary	DOI, embargoes, metrics, institutional agreements	Supports data, code, posters, and presentations; useful for non-traditional outputs
Dataverse	Social sciences	Rich metadata schemas, data citation, access controls	Open-source; deployable institutionally; Harvard Dataverse is the flagship instance

PANGAEA	Earth & environmental science	Georeferenced datasets, ISO metadata, OAI-PMH	Discipline-specific repository with editorial curation

AI-Based and Cloud Computing Tools

The emergence of accessible cloud-based computation and AI-assisted research tools has substantially reduced the hardware barriers faced by researchers in low- and middle-income settings — a structural concern particularly relevant to IGDORE's globally distributed membership (Bezuidenhout et al., 2017). Google Colaboratory provides free GPU-accelerated Jupyter notebook environments, enabling data-intensive analyses without local infrastructure (Bisong, 2019). The Binder project extends this by converting any GitHub repository into a live, executable environment, operationalising the principle of one-click reproducibility (Jupyter Project, 2023).

Large language model (LLM) interfaces — including Claude (Anthropic) and the OpenAI API — are increasingly integrated into scientific workflows for tasks ranging from literature synthesis and code generation to manuscript refinement (Birhane et al., 2023). Critically, their use must be governed by transparent disclosure norms to preserve research integrity (Committee on Publication Ethics [COPE] & STM, 2023). Domain-specific platforms such as the GALAXY Project provide browser-based bioinformatics pipelines that make genomic data analysis reproducible without command-line expertise (The Galaxy Community, 2022). Table 3 maps key AI and cloud tools to their open science applications.

Table 3. AI-based and cloud computing tools for distributed open science.

Tool / Platform	Primary Function	Open Science Application	Access Model
Google Colab	Cloud-based Jupyter notebooks with GPU/TPU	Reproducible data analysis; shareable notebooks	Free tier + Colab Pro
Binder (mybinder.org)	Executable research environments from GitHub repos	One-click reproducibility of published analyses	Free; community-supported
Hugging Face	Model hosting, datasets, inference API	Open ML model sharing and benchmarking	Free and paid tiers
OpenAI / Claude API	Large language model assistance	Literature synthesis, coding support, manuscript drafting assistance	Usage-based pricing
GALAXY Project	Web-based bioinformatics platform	Reproducible genomics workflows without local installation	Free; open-source

OpenCode	Open source AI coding agent for automation and collaborative development	Code generation, collaborative research workflows, and AI-assisted software development	Free; open source, with optional hosted services and support for bring you own key (BYOK) (your own APIs) from external providers

Preprints and Open Peer-Review Platforms

Preprint servers have reshaped the temporal economy of scholarly communication by decoupling the initial disclosure of knowledge from formal peer review (Abdill & Blekhman, 2019). arXiv, founded in 1991, established the model now replicated across disciplines (Ginsparg, 2011). In the life and health sciences, bioRxiv and medRxiv have achieved rapid adoption; studies indicate that preprinted manuscripts attract higher citation rates than non-preprinted counterparts, suggesting visibility benefits beyond speed of dissemination (Fraser et al., 2021).

The open peer-review movement challenges the structural anonymity and opacity of traditional review. Peer Community In (PCI) implements a fully transparent, author-named review system that can function independently of journals, producing citable, publicly accessible recommendation documents (Peer Community In, 2024). F1000Research publishes articles immediately and conducts post-publication open peer review, with all referee reports and author responses permanently visible (F1000Research, 2024). Such models are congruent with IGDORE's commitment to accountability and epistemic transparency. Table 4 provides a structured comparison of these platforms.

Table 4. Preprint servers and open peer-review platforms: models and key features.

Platform	Discipline	Peer Review Model	Key Features
arXiv	Physics, Maths, CS, Economics	Post-publication; community commentary	Oldest and most established preprint server; over 2 million submissions
bioRxiv / medRxiv	Biology / Medicine	Post-publication; links to journal peer review	Widely adopted in life sciences; supports public commenting
PsyArXiv	Psychology & social sciences	Post-publication; OSF-integrated	Key platform for psychological research; strong open data norms
Peer Community In (PCI)	Multidisciplinary	Open, signed peer review before or independent of journals	Referees and recommenders are named; reviews are publicly accessible
eLife	Life & biomedical sciences	Transparent review; referee reports	Pioneering journal-level open peer review model

		published alongside articles	
F1000Research	Multidisciplinary	Post-publication; open peer review with named reviewers	Referee reports and author responses are permanently public

Open Workflow Tools

Reproducible science requires not only that data and outputs be shared, but that the computational and methodological processes generating them be documented and replicable. Version control systems — principally Git, with cloud hosting via GitHub or GitLab — provide a complete, time-stamped audit trail for code and text, supporting the FAIR principles as applied to software (Jimenez et al., 2017). Workflow managers such as Nextflow and Snakemake enable the packaging of multi-step data pipelines as portable, shareable digital objects executable across heterogeneous computing environments (Mölder et al., 2021).

PROTOCOLS.io addresses method transparency by providing versioned, DOI-assigned protocol documentation with community commentary, reducing the replication gap arising from insufficient methodological detail in published papers (Teytelman et al., 2016). Reference management through open tools such as Zotero, which supports group libraries and Citation Style Language (CSL) export, facilitates collaborative bibliography management and open citation practices (Zotero, 2024). The Hypothesis web annotation layer enables persistent, publicly indexable scholarly commentary directly on preprints and journal articles, further extending the open peer commentary ecosystem (Hypothesis, 2024). Table 5 summarises the principal open workflow tools.

Table 5. Open workflow tools: stages, capabilities, and open science relevance.

Tool	Workflow Stage	Core Capability	Open Science Relevance
Git / GitHub / GitLab	Version control & code sharing	Branch-based versioning, issue tracking, CI/CD	Full audit trail; facilitates open-source code publication
Nextflow / Snakemake	Computational pipeline management	Reproducible, portable data analysis pipelines	Enables workflow sharing as FAIR digital objects
PROTOCOLS.io	Methods documentation	Versioned, DOI-assigned step-by-step protocols	Allows living protocols; community commenting and forking
DMP Tool / RDMO	Data management planning	Guided DMP creation aligned with funder templates	Embeds open data commitments from project outset

Zotero	Reference & knowledge management	Group libraries, browser capture, open citation export	Supports collaborative bibliography and open annotation
Hypothesis	Web annotation	Inline public/private annotation on any web document	Enables open scholarly commentary on preprints and articles

Synthesis and Recommended Practices

The described tool landscape showcases what can be achieved with mostly free-to-use, open-source, or community-supported infrastructure. It is essential to note that this serves as a current snapshot, and the tools are evolving rapidly.

For distributed research teams operating within IGDORE's model, the following practices are recommended: (i) adopt platforms that assign persistent identifiers (DOIs) at each stage of the research lifecycle; (ii) prefer open-source or self-hostable tools wherever equivalent functionality exists, to preserve data sovereignty; (iii) pre-register studies on OSF or equivalent registries prior to data collection; (iv) deposit preprints contemporaneously with journal submission; (v) document computational workflows in version-controlled repositories with environment files (e.g., *requirements.txt* or *environment.yml*) to satisfy reproducibility expectations; and (vi) apply Creative Commons licences (preferably CC BY 4.0) to all outputs to maximise downstream reuse (Center for Open Science, 2024; Fecher & Friesike, 2014; Wilkinson et al., 2016).

Collectively, these tools and practices reduce the institutional dependencies that have historically excluded researchers in under-resourced settings, directly supporting the distributed, equity-oriented model of open science championed by IGDORE (Bezuidenhout et al., 2017; Vicente-Saez & Martinez-Fuentes, 2018).

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6 Proof of Concept, Lessons Learned, and Roadmap Toward Large-Scale Implementation

- Integration into COST Action
- Scaling to European consortia
- Long-term sustainability strategy
- Recommendations for funders and programme designers
    - Serge P. J. M. Horbach, Joeri K. Tijdink, Lex Bouter; Research funders should be more transparent: a plea for open applications. R Soc Open Sci. 1 October 2022; 9 (10): 220750. https://doi.org/10.1098/rsos.220750

Integration into COST Action

COST (European Cooperation in Science and Technology) Action is one of the most accessible and structurally compatible funding instruments available to a distributed research network such as IGDORE. COST Actions are interdisciplinary research networks, bringing together researchers and innovators from academia, SMEs, and public institutions to investigate a topic of their choice over a period of four years. The main focus of such Actions is the network and networking activities. Unlike project grants, which typically require a lead institution with substantial administrative infrastructure and demand the immediate delivery of defined outputs, COST Actions fund capacity building, knowledge exchange, training, and the building of a collaborative structure across national boundaries. An institution whose researchers span more than 40 countries and whose defining commitment is to open and replicable science is a natural candidate for participation in this framework.

Integration into a COST Action offers IGDORE-affiliated researchers several concrete benefits. It provides access to funding for networking activities, such as workshops, short-term scientific missions, conference attendance, and dissemination, without requiring the kind of institutional overhead that disadvantages smaller or independent research organizations in conventional grant competitions.

In practice, the easiest way to integrate IGDORE-affiliated members into COST Actions is to identify running Actions whose thematic focus overlaps with IGDORE's goals of promoting open science, open source, and transparent reporting. Although these are central values of European research, upon reviewing the current Actions, we were unable to pinpoint any such Actions; therefore, **we have decided to apply for a new COST Action**. This approach has several advantages: an Action that closely aligns with IGDORE's objectives provides

formal recognition within the European research landscape; it also enhances the credibility of affiliated researchers when seeking additional funding or collaborative partnerships. Last, but not least, this can serve as a foundation for future grant applications and create structured opportunities to connect IGDORE's methodological commitments with the broader European open science agenda.

Scaling to European Consortia

COST Actions have a known high follow-up success rate: statistically, 39% of proposals submitted as a follow-up of a COST Action are funded. Therefore, one long-term goal of participation in COST Actions is a preparatory step toward engagement with larger European research consortia under Horizon Europe programmes. The capacity requirements for consortium participation at this scale are substantially greater: project management, financial reporting, and the coordination of multi-partner deliverables demand administrative infrastructure that independent researchers working in isolation cannot readily provide. IGDORE's role in this context will be to function as the institutional anchor that makes consortium participation viable for researchers who would otherwise be unable to access these funding streams.

Several features of IGDORE's model are relevant to this scaling ambition. The institute's location independence and multi-national membership mean that it can, in principle, fulfill the partner diversity requirements of European consortia without the geographic constraints that limit single-country institutions. However, this location-independent structure does not correspond to the current scientific model. The explicit commitment to open science practices aligns directly with the open science conditions embedded in Horizon Europe funding agreements, reducing the compliance burden that some partners find onerous.

IGDORE's current model, which relies on Swedish grant management infrastructure, will require significant development to support the volume and complexity of activity that meaningful Horizon Europe participation entails. Taking this step requires anticipatory planning rather than reactive adaptation.

Long-Term Sustainability Strategy

The financial sustainability of an independent research institute — operating without an endowment or government funding —is a natural structural challenge. IGDORE's current Swedish NGO model relies solely on external donor support. There are no membership-based affiliations, and grant management fees are also in early stages. Hence, IGDORE is lacking operational flexibility and does not, in itself, constitute a stable, long-term funding base. A sustainable strategy would require diversification across the following three dimensions.

- **Revenue diversification**: reducing dependence on any single source by developing complementary income streams, including, where appropriate, fee-based services — methodological consultancy, open science training, research integrity support — that draw on the expertise of affiliated researchers and serve an identifiable external demand.
- **Increase grant management capacity**: Expanding beyond the current Swedish-based infrastructure to access complementary non-dilutive funding arrangements (grants) will broaden opportunities for affiliated researchers and reduce the risk of dependency on a single source.

- **Community diversification**: growing the affiliated researcher base in a deliberate and quality-controlled manner, such that the institute's collective intellectual capacity and network reach increase without compromising the standards of conduct that define its value proposition.

Sustainability also has a reputational dimension. An institute whose affiliated researchers consistently produce work of demonstrable methodological rigor, publish in credible venues, and actively engage with the open science community accumulates a form of institutional capital that holds value in itself and is not easily replicated. This capital will provide a stable foundation for future funding in the form of grant applications, partnership negotiations, and donor engagement. In this respect, the long-term sustainability strategy is inseparable from the quality assurance and community standards discussed elsewhere in this white paper.

Recommendations for Funders and Programme Designers

The funding landscape for distributed, independent research remains largely unrecognized at present. Existing grant programmes — whether national, European, or international — have been designed around assumptions of institutional concentration: a lead university, a defined research team in a fixed location, overhead structures calibrated to large administrative bodies. IGDORE's unique model is hard to fit into these templates, and to date, we are still looking for instruments that are well aligned to support remote and distributed research as an organizational form in its own right.

This lack of funding is not simply a gap in provision. The distributed research model represented by IGDORE is still experimental in many aspects. Questions remain about how such a community can sustain methodological standards without physical co-location, maintain institutional coherence across national and disciplinary boundaries, and deliver research outputs of a quality comparable to, or even better than, those of conventionally organized institutions. This whitepaper is also intended to demonstrate the model's viability and to support the case for its formal recognition by the funding community.

The recommendation to funders and programme designers is therefore not a request to accommodate an established and well-documented practice. It is rather an invitation to participate in a structured experiment. Funding instruments that pilot support for distributed research networks — with appropriate monitoring, evaluation, and reporting frameworks — would serve two purposes simultaneously: they would provide the resources necessary for the model to develop and be tested at a meaningful scale, and they would generate the evidence that future funding decisions, by this and other agencies, will require.

Specifically, this involves developing eligibility criteria and administrative requirements that do not presuppose significant institutional infrastructure, as well as evaluation frameworks that directly assess research quality and methodological rigor rather than inferring them from institutional prestige. It also requires a willingness to treat organizational innovation in research as a legitimate investment opportunity, subject to the same standards of transparency and accountability as scientific outputs.

The open applications framework proposed by Horbach, Tijdink, and Bouter (2022) — in which grant applications, review reports, and funding decisions are made publicly available — would be a particularly well-suited model for any pilot programme of this kind, given that transparency and reproducibility of process are foundational commitments of the research community it would serve.

The absence of precedent is not, in this context, an argument for caution. It is an argument for deliberate, well-documented action.

7 Public outreach

Importance of Open Science

In today's research landscape, the importance of open science cannot be overlooked. It promotes broader access to knowledge, enhances systemic transparency, fosters collaboration, accelerates innovation, and response during (global) crises. The major benefits that open science provides at present can be summarized as:

- **Balanced participation across the global scientific community:** By making knowledge openly available, individuals from diverse backgrounds, including students, independent researchers, and policymakers, can access and utilize information more equitably.
- **Increased Productivity and Innovation:** Sharing data, code, and results allows others to build on existing work instead of repeating it. This speeds up discovery, prevents mistakes from spreading, and keeps research focused on new breakthroughs.
- **Practical solution to public skepticism:** When the internal processes of research are made visible and verifiable, it becomes significantly easier for both the public and the scientific community to trust the outcomes.
- **Dissemination of findings in accessible formats**: Presenting data in easy-to-interpret ways enables communities and decision-makers to bridge the gap between theoretical knowledge and real-world application.
- **Crisis response**: During periods of crisis, such as natural disasters or climate-related emergencies, immediate access to reliable information ensures that response efforts are guided by the best available evidence.
- **Enhanced Accessibility for Non-Specialists**: Open science should extend beyond making data available; it must aim to make research understandable through plain-language summaries, local-language communication, and visual materials. This ensures that findings are useful for community engagement, schools, and non-specialist audiences.

Current state

The landscape of open science is rapidly evolving as it moves from a theoretical ideal to a global standard. This transformation is driven by several key developments:

- **Global Adoption and Policy through international frameworks:** Major organizations, such as UNESCO and various national governments, are now implementing formal mandates requiring research outputs to be publicly accessible.
- **Funding from both public and private sectors:** Grant-making bodies both in public and private sectors are increasingly requiring "Open Access" as a condition for financial support.

- **Rise of Citizen Science:** There is a growing movement toward "Citizen Science," which encourages active collaboration between researchers and the general public, allowing for larger datasets and more diverse perspectives.
- **Cultural Shift within academic institutions**: There is a growing movement to reform traditional "publish or perish" metrics, rewarding researchers for data sharing and collaboration rather than just high-impact journal placements.
- **Infrastructure Support to sustain long-term accessibility:** The growth of digital repositories, interoperable data standards, and cloud-based collaborative platforms provides the technical foundation necessary for global knowledge exchange.
- **Journalistic Integration:** There is a growing trend in "Data Journalism," in which researchers collaborate directly with major news outlets to create interactive maps and evidence-based stories.

Practical steps towards Open Science

Transitioning to an open research model requires active participation through various digital and community-driven channels. There are several steps involved in contributing to and transitioning towards open science, including:

- **Engaging Citizen Science in data collection and analysis:** This democratizes the research process and allows for larger-scale observations that would be impossible for a single team to manage.
- **Using Open Repositories:** Archiving and sharing datasets, preprints, and supplemental materials on platforms like Zenodo or OSF ensures that scholarly works are findable, accessible, interoperable, and reusable.
- **Issuing Strategic Press Releases:** Partnering with institutional communications offices to distribute "Research Briefs" to local and international media ensures that open research findings are picked up by news desks and policy influencers, increasing public reach.
- **Contributing to Op-Eds and Feature Articles:** Writing for national and international newspapers or digital news platforms helps bridge the gap between technical data and the public understanding of science, making research more relatable to society.
- **Utilizing Global Knowledge Hubs and Specialized Platforms:** Sharing technical reports, briefings, and policy papers on high-traffic repositories and specialized platforms—such as **ReliefWeb** for humanitarian research or **PreventionWeb** for risk-related studies—ensures that critical information reaches practitioners and decision-makers in real-time.
- **Participating in Open Access Publishing:** Choosing journals that make findings freely available to everyone removes financial barriers for readers and increases global visibility.
- **Participating in Open Science Events:** Forums such as "hackathons," workshops, and unconferences foster networking and the exchange of best practices across different scientific disciplines.
- **Using Open Protocols to document experimental methods:** Sharing detailed methodology for research helps reduce waste and prevents other labs from repeating avoidable mistakes.

- **Registering for an ORCID ID:** This unique persistent identifier will link all the research outputs together, regardless of name changes or institutional moves.
- **Securing Dedicated Outreach Funding:** Budgeting specifically for "Science Communication" and "Public Engagement" within grant applications to cover the costs of Article Processing Charges (APCs), professional translation into local languages, and the creation of non-technical visual materials for non-specialist audiences.

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Appendix 1. Collaboration Charter

Purpose of this collaboration

This white paper is:

1. A conceptual framework for distributed open science, and
2. A live proof-of-concept (PoC) demonstrating distributed research collaboration in practice.

The goal is to co-create a high-quality, publishable white paper while testing governance structures, digital tools, and collaboration workflows suitable for large-scale international interdisciplinary research networks. Therefore, further purposes include expanding horizons of knowledge and understanding and scientific inquiry for more equitable inclusion; cross-pollination/learning across disciplines; implementing processes and practices of Open Research that actualize principles of the collaborative aims we speak of; and identifying the current challenges of Open Research with the intention to outline future research and development directions.

This collaboration aims to be:

- Transparent
- Inclusive
- Efficient
- Documented
- Reproducible

2. Scope of Work

The white paper will include:

- Conceptual framework
- Best practices
- Tool ecosystem
- Governance models
- Roadmap toward implementation

Out of scope:

- Empirical primary research
- Full policy drafting
- Technology development beyond testing and documentation

3. Roles and Responsibilities

3.1 Core Roles

Scientific Lead

- Oversees coherence and final quality.
- Has final editorial responsibility.

Technical Lead

- Responsible for the thematic modules.
- Coordinate contributors within their section.

- Deliver draft within agreed timeline.

Contributors

- Draft content, comment, and revise.
- Participate in at least cross-review rounds.

4. Governance and Decision-Making

4.1 Structure Freeze Principle

The structure of the document will be agreed at kickoff and only modified by consensus of the participants. A suggested structure can be found at the end of this document.

4.2 Decision-Making

- **Minor decisions**: Section-level agreement.
- **Cross-cutting decisions**: Technical Lead + Scientific Lead.
- In the event of an **unresolved disagreement**, the Scientific Lead decides, with the rationale documented.

4.3 Transparency

All major decisions will be logged in a shared decision log.

5. Authorship and Contribution Recognition

Authorship will follow transparent criteria:

Authorship and contribution recognition will be defined in accordance with the principles of transparency, accountability, and integrity as recommended by the Committee on Publication Ethics (COPE). Authorship will reflect only those individuals who meet established criteria for substantial intellectual contribution, and all contributors will be appropriately credited based on their role.

Lead Authors

Lead authors are individuals who take primary responsibility for the integrity of the work as a whole. They are expected to have made substantial contributions to the conception or design of the study, the development of methodology, data analysis or interpretation, and the drafting or critical revision of the manuscript. Lead authors also assume responsibility for ensuring that all co-authors meet authorship criteria and that the final version is approved by all contributors.

- Scientific lead and Technical lead with substantial intellectual contribution.

Co-authors

Co-authors are individuals who have made significant intellectual contributions to the work in accordance with COPE authorship standards. This may include contributions to study design, data acquisition, formal analysis, interpretation of results, or substantial involvement in drafting, revising, or approving the manuscript. All co-authors are expected to agree to be accountable for their contributions and for the integrity of the work.

- Contributors with substantive writing, conceptual, or analytical input.

Acknowledgements

- Participants who provided feedback, comments, or minor edits.

Contribution categories may be documented using a simple contribution matrix (e.g., conceptualization, drafting, review, coordination).

No honorary authorship.

Contribution Documentation

To ensure full transparency and traceability of contributions, author roles will be documented using the CRediT (Contributor Roles Taxonomy). This may include, but is not limited to: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing (original draft preparation), writing (review and editing), visualization, supervision, project administration, and funding acquisition.

6. Collaboration Infrastructure

The PoC will use:

- A shared writing platform (e.g., Google Docs or Git-based repository)
- A communication channel (e.g., forum "On Science and Academia" or Slack)
- A decision log
- A shared folder for references and templates

All drafts will be version-controlled.

7. Timeline and Workflow

- Phase 1 – Kickoff & Structure Freeze
- Phase 2 – Parallel Section Drafting
- Phase 3 – Cross-Review
- Phase 4 – Harmonisation
- Phase 5 – Meta-Reflection & Finalisation

Detailed timeline will be agreed at kickoff.

8. Inclusiveness and Conduct

The collaboration commits to:

- Respectful and constructive dialogue
- Gender-balanced and geographically diverse participation
- Clear and accessible language
- Equal opportunity to contribute across career stages

Disagreements are expected and will be handled respectfully and transparently.

9. Evaluation of the PoC

At the end of the writing process, participants will complete a short feedback survey covering:

- Tool usability
- Governance clarity
- Coordination efficiency
- Inclusiveness experience
- Improvement suggestions

These insights will be included in the final white paper.

10. Sustainability and Future Use

Outputs of this collaboration may inform:

- COST Action proposal
- Future Horizon proposals
- IGDORE governance framework
- Public policy discussions

All contributors agree that the white paper will be shared openly under an agreed license.

Endorsement

By participating in this PoC, contributors agree to follow this Collaboration Charter and support transparent, inclusive, and distributed research practices.